Real field (实数域)

Defn (ε-δ-definition) Suppose that $x_0 \in (a,b)$ and $(a,b)\setminus\{x_0\} \subseteq \text{dom } f$, we have $\lim\limits_{x\to x_0} f(x) = A$ if and only if for any $\varepsilon > 0$, there exists $\delta > 0$, such that

$$|f(x) - A| < \varepsilon \quad \text{whenever} \quad 0 < |x - x_0| < \delta.$$

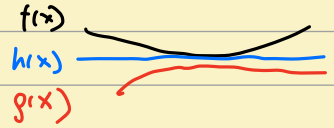
Defn ($x \to +\infty$) Suppose that the domain of f contains $(\alpha, +\infty)$ for some $\alpha \in \mathbb{R}$, we have $\lim\limits_{x\to+\infty} f(x) = A$ if and only if for any $\varepsilon > 0$, there exists $K > \alpha$, such that

$$|f(x) - A| < \varepsilon, \quad \text{whenever} \quad x > K$$

Similarly, we can define $\lim\limits_{x\to-\infty} f(x) = A$, $\lim\limits_{|x|\to+\infty} f(x) = A$

Thm (Sandwich theorem) Suppose that $0 < |x - x_0| < \delta$, if

$$f(x) \le h(x) \le g(x)$$



and $\lim\limits_{x\to x_0} f(x) = \lim\limits_{x\to x_0} g(x) = A$, then we have

$$\lim_{x\to x_0} h(x) = A$$

Thm (Two important limits)

① $\lim\limits_{x\to 0} \dfrac{\sin x}{x} = 1$ ② $\lim\limits_{x\to+\infty} \left(1+\dfrac{1}{x}\right)^x = e$

Ex 2

$\lim\limits_{n\to\infty} \left(1+\frac{1}{n}\right)^n = e$

$n \in \mathbb{Z}_+$

Proof: Let $x \ge 1$, then

$$\left(1+\frac{1}{[x]+1}\right)^{[x]} \le \left(1+\frac{1}{x}\right)^x \le \left(1+\frac{1}{[x]}\right)^{[x]+1}$$

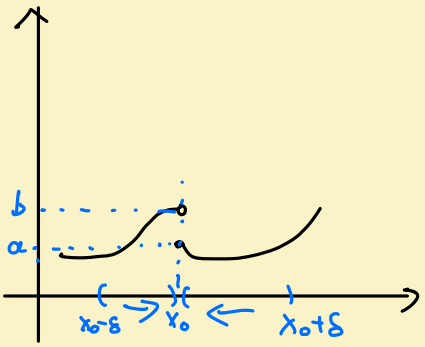
$$\Rightarrow \left(1+\frac{1}{[x]+1}\right)^{[x]+1}\left(1+\frac{1}{[x]+1}\right)^{-1} \le \left(1+\frac{1}{x}\right)^x \le \left(1+\frac{1}{[x]}\right)^{[x]}\cdot\left(1+\frac{1}{[x]}\right)$$

$$\lim_{[x]+1\to+\infty}\left(1+\frac{1}{[x]+1}\right)^{[x]+1}\left(1+\frac{1}{[x]+1}\right)^{-1}=e\cdot 1$$

$$\lim_{[x]\to+\infty}\left(1+\frac{1}{[x]}\right)^{[x]}\cdot\left(1+\frac{1}{[x]}\right)=e\cdot 1$$

Therefore $\lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^{x}=e$

One-side limits

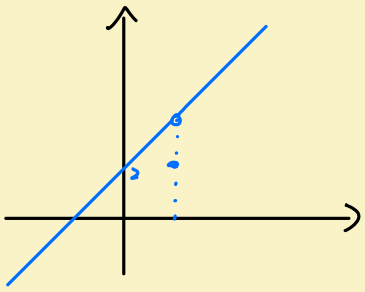


Defn (One-side limit) Suppose that $(x_0-\alpha, x_0)\subset \text{dom}\, f$ for some $\alpha>0$, we have $\lim_{x\to x_0^-} f(x)=A$ if and only if for any $\varepsilon>0$, there exists $0<\delta<\alpha$, such that

$$|f(x)-A|<\varepsilon \quad \text{whenever} \quad x\in(x_0-\delta, x_0)$$

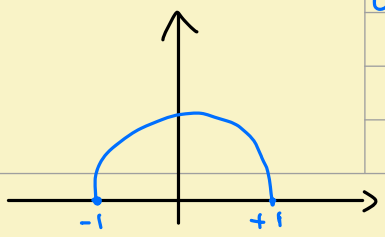
Defn (One-side limit) Suppose that $(x_0, x_0+\alpha)\subset \text{dom}\, f$ for some $\alpha>0$, we have $\lim_{x\to x_0^+} f(x)=A$ if and only if for any $\varepsilon>0$, there exists $0<\delta<\alpha$, such that

$$|f(x)-A|<\varepsilon \quad \text{whenever} \quad x\in(x_0, x_0+\delta)$$



Thm: $\lim_{x\to x_0} f(x)=A$ if and only if $\lim_{x\to x_0^-} f(x)=\lim_{x\to x_0^+} f(x)=A$

E.g.: $f(x)=\begin{cases}\frac{x^2-4}{x-2} & x\neq 2\\ 2 & x=2\end{cases}$ $\lim_{x\to 2} f(x)=4\neq f(2)$



E.g.: $f(x)=\sqrt{1-x^2}$ consider $\lim_{x\to 1^-} f(x)$ and $\lim_{x\to -1^+} f(x)$

$\lim_{x\to 1^-} f(x)=0$. since $|x|\le 1$, we have $1-x^2=(1+x)(1-x)\le 2(1-x)$

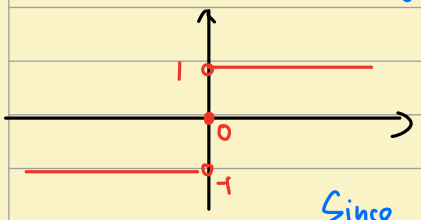
for any $0<\varepsilon<1$, we can find x such that

$2(1-x) < \varepsilon^2 \Rightarrow \sqrt{1-x^2} < \frac{\varepsilon}{\sqrt{2}} < \varepsilon$

take $\delta = \frac{\varepsilon^2}{2}$, then $0 < 1-x < \delta \Rightarrow x \in (1-\delta, 1)$

we have $|\sqrt{1-x^2} - 0| < \varepsilon$ that is $\lim_{x \to 1^-} \sqrt{1-x^2} = 0$

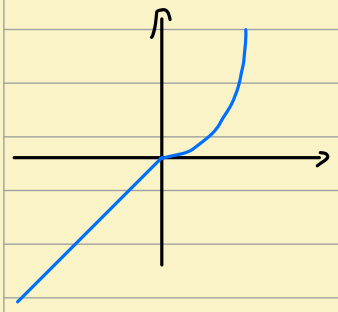
E.g. $f(x) = \operatorname{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$ $\lim_{x \to 0^-} f(x)$, $\lim_{x \to 0^+} f(x)$, $\lim_{x \to 0} f(x)$



$\lim_{x \to 0^-} f(x) = -1$ $\lim_{x \to 0^+} f(x) = 1$

Since $\lim_{x \to 0^-} f(x) \neq \lim_{x \to 0^+} f(x)$, the limit $\lim_{x \to 0} f(x)$ does not exist.

E.g. $f(x) = \begin{cases} x^2 & x \geq 0 \\ x & < 0 \end{cases}$ verify $\lim_{x \to 0} f(x)$



$\lim_{x \to 0^-} f(x) = 0$ $\lim_{x \to 0^+} f(x) = 0$

$\lim_{x \to 0} f(x) = \lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x) = 0$

Six types of limits

$\lim_{x \to x_0} f(x)$ $\lim_{x \to x_0^-} f(x)$ $\lim_{x \to x_0^+} f(x)$ $\lim_{x \to +\infty} f(x)$ $\lim_{x \to -\infty} f(x)$ $\lim_{|x| \to +\infty} f(x)$

Properties of limits

① Uniqueness (唯一性) If $\lim_{x\to x_0} f(x)=a$ and $\lim_{x\to x_0} f(x)=b$, then $a=b$.

② Boundedness (有界性) If $\lim_{x\to x_0} f(x)$ exists, then $f(x)$ is bounded at $0<|x-x_0|<\delta$

If $\lim_{x\to\infty} f(x)$ exists, then $f(x)$ is bounded at $x>k$

⋮

E.g. $\lim_{x\to x_0} f(x)=+\infty$ ← the limit of f at x_0 does not exist.

③ (保号性) If $\lim_{x\to x_0} f(x)=A>0$, then for any $0<r<A$, we have $0<r<f(x)$ whenever $0<|x-x_0|<\delta$.

④ (保不等式性) If $\lim_{x\to x_0} f(x)$ and $\lim_{x\to x_0} g(x)$ exist, and $f(x)\le g(x)$ whenever $0<|x-x_0|<\delta$, then

$$\lim_{x\to x_0} f(x) \le \lim_{x\to x_0} g(x)$$

Proof: ① Suppose $\lim_{x\to x_0} f(x)=a$, and $\lim_{x\to x_0} f(x)=b$. By definition

$\forall \varepsilon>0$, $\exists \delta_1>0$, such that $0<|x-x_0|<\delta_1 \Rightarrow |f(x)-a|<\frac{\varepsilon}{2}$

$\forall \varepsilon>0$, $\exists \delta_2>0$, such that $0<|x-x_0|<\delta_2 \Rightarrow |f(x)-b|<\frac{\varepsilon}{2}$

$$|a-b| = |(f(x)-a)-(f(x)-b)|$$
$$\le |f(x)-a| + |f(x)-b| < \frac{\varepsilon}{2}+\frac{\varepsilon}{2} < \varepsilon$$

then we have $a=b$

② $\forall \varepsilon>0$, $\exists \delta>0$, such that $0<|x-x_0|<\delta \Rightarrow |f(x)-A|<\varepsilon$

$$|f(x)| = |f(x)-A+A| \le |f(x)-A| + |A| < \varepsilon + |A|$$

$0<r<A$

$\Rightarrow 0<r<f(x)$

$|f(x)-A|<\varepsilon$

$\Rightarrow -\varepsilon < f(x)-A < \varepsilon$

$\Rightarrow A-\varepsilon < f(x)$

$f(x) < \varepsilon + A$

Hence $f(x)$ is bounded on $(x_0-\delta, x_0)\cup(x_0, x_0+\delta)$

③ Suppose $A>0$, for any $0<r<A$, take $\varepsilon = A-r$, by definition

there exists $\delta>0$, $0<|x-x_0|<\delta \Rightarrow |f(x)-A|<\varepsilon$

$\Rightarrow f(x) > A-\varepsilon = r$

④ Suppose $f(x)\le g(x)$ holds for any $x\in(x_0-\delta, x_0)\cup(x_0, x_0+\delta)$

By definition:

$\forall \varepsilon>0$, $\exists \delta_1>0$, such that $0<|x-x_0|<\delta_1$

$\Rightarrow |f(x)-A|<\varepsilon$, $\Rightarrow f(x) > A-\varepsilon$

$\forall \varepsilon>0$, $\exists \delta_2>0$, such that $0<|x-x_0|<\delta_2$

$\Rightarrow |g(x)-B|<\varepsilon$, $\Rightarrow g(x) < B+\varepsilon$

Take $\delta = \min(\delta_1, \delta_2)$, when $0<|x-x_0|<\delta$,

$A-\varepsilon < f(x) \le g(x) < B+\varepsilon$

$\Rightarrow A < B+2\varepsilon \quad \Rightarrow A \le B$

Operations of limits

Prop: (Arithmetic operations)

a) $\lim c\cdot f(x) = c\cdot \lim f(x)$ c is a constant

b) $\lim (f(x)\pm g(x)) = \lim f(x) \pm \lim g(x)$

c) $\lim f(x)\cdot g(x) = \lim f(x)\cdot \lim g(x)$

d) $\lim \frac{f(x)}{g(x)} = \frac{\lim f(x)}{\lim g(x)}$ given $g(x)\neq 0$ and $\lim g(x)\neq 0$

Prop: a) $\lim (f(x))^n = (\lim f(x))^n$

b) $\lim c = c$

c) $\lim \sqrt[n]{f(x)} = \sqrt[n]{\lim f(x)}$

Thm Suppose $\lim_{y\to y_0} f(y) = A$, $\lim_{x\to x_0} g(x) = y_0$ then $\lim_{x\to x_0} f(g(x)) = A$

Proof: By definition

$\forall \varepsilon_1 > 0$, $\exists \delta_1 > 0$, such that $0 < |y - y_0| < \delta_1 \Rightarrow |f(y) - A| < \varepsilon_1$

$\forall \varepsilon_2 > 0$, $\exists \delta_2 > 0$, such that $0 < |x - x_0| < \delta_2 \Rightarrow |g(x) - y_0| < \varepsilon_2$

Take $\delta_1 = \varepsilon_2$

$\Rightarrow \forall \varepsilon_1 > 0$ $\exists \delta_2 > 0$, such that $0 < |x - x_0| < \delta_2 \Rightarrow |g(x) - y_0| < \varepsilon_2$

Since $\delta_1 = \varepsilon_2 \Rightarrow |f(g(x)) - A| < \varepsilon_1$

By definition $\lim_{x\to x_0} f(g(x)) = A$

E.g.: find $\lim_{x\to -2} \frac{x^3 + 2x^2 - 1}{5 - 3x} = \lim_{x\to -2} \frac{(-2)^3 + 2(-2)^2 - 1}{5 - 3(-2)} = -\frac{1}{11}$

E.g.: $\lim_{x\to 1} \frac{x^2 - 1}{x - 1} = \lim_{x\to 1} \frac{(x-1)(x+1)}{x-1} = \lim_{x\to 1} x + 1 = 2$

$\lim_{x\to 0} \frac{\sqrt{x^2+9} - 3}{x^2} = \lim_{x\to 0} \frac{(\sqrt{x^2+9} - 3)(\sqrt{x^2+9} + 3)}{x^2(\sqrt{x^2+9} + 3)} = \lim_{x\to 0} \frac{x^2}{x^2} \cdot \frac{1}{\sqrt{x^2+9} + 3} = \frac{1}{6}$

$\lim_{x\to +\infty} \frac{3x^2 + 2x - 1}{x^2 - 2} = \lim_{x\to +\infty} \frac{3 + \frac{2}{x} - \frac{1}{x^2}}{1 - \frac{2}{x^2}} = 3$

E.g. Suppose $\lim_{x\to x_0} f(x) = A$, prove $\lim_{h\to 0} f(x_0 + h) = A$

Proof: Let $x = x_0 + h$, then $h = x - x_0$, we have $x \to x_0$ as $h \to 0$

$\lim_{x\to x_0} f(x) = A \iff \lim_{h\to 0} f(x_0 + h) = A$

($x = x_0 + h$ above the left limit; $h \to 0$ below the left limit)

E.g.: Since $\lim_{x\to+\infty}(1+\frac{1}{x})^x = e$, prove $\lim_{x\to-\infty}(1+\frac{1}{x})^x = e$

Proof: Let $y = -(x+1)$ then $x = -y-1$ $x\to-\infty$ as $y\to+\infty$

$$\lim_{x\to-\infty}(1+\frac{1}{x})^x = \lim_{y\to+\infty}(1-\frac{1}{y+1})^{-(y+1)} = \lim_{y\to+\infty}\left[\left(\frac{y}{y+1}\right)^{-1}\right]^{y+1}$$

$$= \lim_{y\to+\infty}\left(\frac{y+1}{y}\right)^{y+1} = \lim_{y\to+\infty}(1+\frac{1}{y})^{y+1} = \lim_{y\to+\infty}(1+\frac{1}{y})^y\cdot(1+\frac{1}{y}) = e$$

E.x. Prove $\lim_{x\to0}(1+x)^{\frac{1}{x}} = e$

Proof: $y = \frac{1}{x}$, then $x = \frac{1}{y}$, $x\to0$ as $y\to\pm\infty$

since $\lim_{y\to+\infty}(1+\frac{1}{y})^y = \lim_{y\to-\infty}(1+\frac{1}{y})^y = e$

we have $\lim_{x\to0}(1+x)^{\frac{1}{x}} = \lim_{|y|\to+\infty}(1+\frac{1}{y})^y = e$

Note: Heine's theorem provides a linkage between the limit of function and the limit of sequence.

Thm (Heine's theorem) $\lim_{x\to x_0} f(x) = A$ if and only if for any sequence, $\{x_n\}\subset \text{dom } f$ satisfying $x_n\neq x_0$, $x_n\to x_0$ as $n\to+\infty$, we have $\lim_{n\to+\infty} f(x_n) = A$

Proof: ($\Rightarrow$) Suppose $\lim_{x\to x_0} f(x) = A$, then for any $\varepsilon>0$, $\exists\delta>0$, such that $0<|x-x_0|<\delta \Rightarrow |f(x)-A|<\varepsilon$. Let $\{x_n\}\subset(x_0-\delta,x_0)\cup(x_0,x_0+\delta)$ and $\lim_{n\to+\infty} x_n\to x_0$, then there exists $N\in\mathbb{N}$ such that $0<|x_n-x_0|<\delta$, therefore $|f(x_n)-A|<\varepsilon \Rightarrow \lim_{n\to\infty} f(x_n) = A$.

($\Leftarrow$) Suppose $\{x_n\}\subset(x_0-\delta,x_0)\cup(x_0,x_0+\delta)$ and $x_n\to x_0$ as $n\to+\infty$, we have $\lim_{n\to\infty} f(x_n) = A$. Prove by contradiction, suppose that $\lim_{x\to x_0} f(x)\neq A$

then $\exists \varepsilon_0 > 0$, $\forall \delta > 0$, $|f(x) - A| \geq \varepsilon_0$ whenever $0 < |x - x_0| < \delta$.

Take $\delta' = \delta, \frac{\delta}{2}, \frac{\delta}{3}, \cdots \frac{\delta}{n}$ there exists corresponding $x_1, x_2 \cdots x_n$ such that $0 < |x_n - x_0| < \frac{\delta}{n}$, but $|f(x_n) - A| \geq \varepsilon_0$

But $\{x_n\} \subset (x_0 - \delta, x_0) \cup (x_0, x_0 + \delta)$, $x_n \to x_0$ as $n \to +\infty$, but $|f(x_n) - A| \geq \varepsilon_0$. contradiction. Therefore $\lim_{x \to x_0} f(x) = A$

Thm (Heine's theorem, alternative)

$\lim_{x \to x_0} f(x) = A \iff \forall x_n \to x_0$, we have $\lim_{n \to +\infty} f(x_n) = A$.

E.g.: Prove $\lim_{x \to 0} \sin \frac{1}{x}$ does not exist.

Proof: Let $x_n = \frac{1}{2n\pi}$, $y_n = \frac{1}{2n\pi + \frac{\pi}{2}}$ then we have

$x_n \to 0$, $y_n \to 0$ as $n \to +\infty$, but $\sin \frac{1}{x_n} = \sin 2n\pi = 0$

$\sin \frac{1}{y_n} = \sin\left(2n\pi + \frac{\pi}{2}\right) = 1$ for any $n \in \mathbb{N}$.

Let $z_n = \begin{cases} x_{2n} & n \text{ is odd} \\ y_{2n} & n \text{ is even} \end{cases}$ then $\lim_{n \to +\infty} \sin \frac{1}{2n}$ is not convergent.

Then $\lim_{x \to 0} \sin \frac{1}{x}$ does not exist.